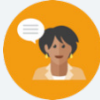


13.6 Exploring Transformations – Vertical Shifts

Lesson Introduction

 Benchmark	MA.912.F.2.1 Identify the effect on the graph or table of a given function after replacing $f(x)$ by $f(x) + k$, $kf(x)$, $f(kx)$, and $f(x + k)$ for specific values of k .		 Learning Targets	- I can discover the effect on the graph of $f(x)$ for a transformation of $f(x + k)$ or $f(x) + k$. - I can discover the effect on a table of values of $f(x)$ for a transformation of $f(x + k)$ or $f(x) + k$.
 Benchmark Coherence	Building On N/A		Working Toward N/A	
 Essential Questions	- How can I determine the effect on the graph of $f(x)$ for a transformation of $f(x) + k$? - How can I determine the effect on a table of values of $f(x)$ for a transformation of $f(x) + k$?			
 Lesson Narrative	In this lesson, students continue their learning of transformations. Students will begin by exploring vertical transformations of the graphs of functions. They will then bring together their learning from the Exploration and the prior lesson to describe the effect on the graph of functions that have been transformed by a vertical or horizontal shift. They will conclude the lesson by practicing transformations of functions with both vertical and horizontal shifts.			
 Component Summary	13.6.1 Warm-Up (~5 min.)	Bell Work with horizontal shifts (<i>SE p. 263</i>)	13.6.3 Bring It Together (10 min.)	Vertical and horizontal transformations table activity (<i>SE p. 266</i>)
	13.6.2 Exploration (10-15 min.)	Vertical transformations exploration activity (<i>SE p. 264</i>)	13.6.4 Guided Practice (10-15 min.)	Transformations practice activity (<i>SE p. 266</i>)
	13.6.5 Wrap-Up (~5 min.)	Growth Mindset activity (<i>SE p. 267</i>)		
 Resources	Glossary		Materials	
	<u>Key Terms:</u> N/A <u>Additional Terms:</u> N/A		Desmos parent functions.	
		Additional Preparation Devices to run graphing software.		

 Teacher Tips	Common Misconceptions		Things to Consider	
	- Students may interpret the negative sign as a vertical translation up the y-axis. - Students may interpret the vertical shift as a horizontal shift when provided with the function.		13.6.2 Exploration, students are exploring shifting the graph of a function. There are points where students may get confused and struggle. Be cautious not to give away anything during the Exploration, but instead, allow the Exploration to unfold naturally as designed.	
	Literacy and Language Routines		Mathematical Thinking and Reasoning (MTR) Standard	
 Differentiate Instruction	Think-Pair-Share 13.6.2 Exploration is an exploration of vertical transformations that can be done with a partner. Consider leveraging this routine to allow students to try pieces on their own and then share with their partner to calibrate and, if necessary, come to consensus on the correct answer.		MA.K12.MTR.4.1: Discussion and Reflection In 13.6.2 Exploration, students will work with their partner to complete the Exploration task. Part of the Exploration outcomes are for students to explore concepts to make formalized connections with a future activity. During this activity, students will justify results by explaining methods and processes and construct possible arguments based on evidence.	
	Consider the use of digital representations for students to explore as part of this lesson. For example, use Desmos with sliders for students to visualize transformations.			
	Lesson Notes:			

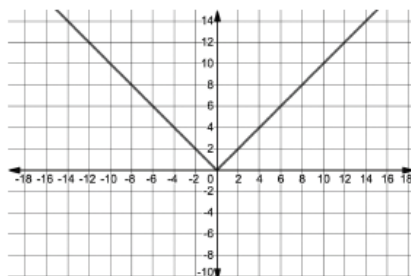
13.6.1 Warm-Up: Bell Work

Component Overview: Students will complete two horizontal-shift problems as Bell Work.

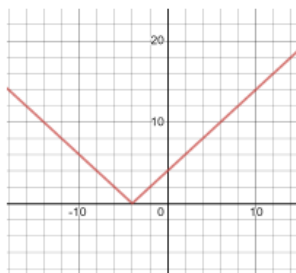


13.6.1 Warm-Up: Bell Work

- The graph of the function, $G(x)$, is shown on the coordinate grid.



Draw or describe the graph of $G(x + 4)$.



The graph shifted four units to the left compared to the original.

- The table of values of the function, $y = H(x)$, is shown.

x	-8	-6	-4	0	2	6	10
H	6	3	0	-6	-3	-9	-15

Describe the effect on x -values for the function

$$y = H(x - 5).$$

The x -values would have an increase of 5.



This activity assesses what students recall about horizontal shifts, which were covered in the previous lesson. Use this data to help determine which students may still be struggling with that type of shift. Horizontal shifts will be part of this lesson, so this data can be a valuable insight into what misconceptions students have to be able to address as part of this lesson.

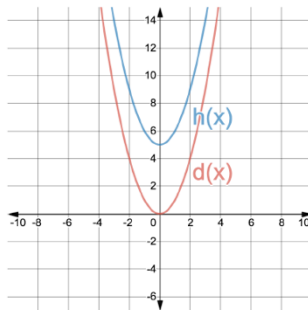
13.6.2 Exploration: Vertical Transformations

Component Overview: Students will explore vertical transformations with seven problems. For this activity, students will need a partner to complete.



13.6.2 Exploration: Vertical Transformations

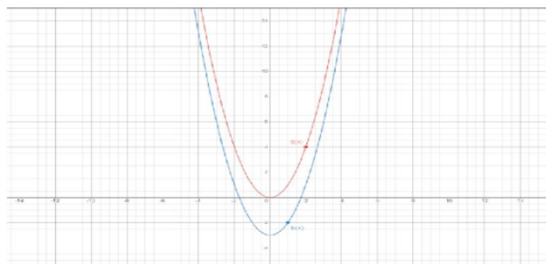
The graphs of $y = d(x)$ and $y = h(x)$ are shown.



1. When a graph was shifted left or right, the x -values were affected. Discuss with your partner which values are affected by shifting the graph of $y = d(x)$ up 5 units to create $y = h(x)$. Summarize your discussion.

The y -value is increased by 5 units.

2. On the coordinate plane, draw the graph of the function $k(x)$ by shifting $d(x)$ down 3 units.



Red graph is $d(x)$ and it shifted three units down resulting in the blue graph of $k(x)$.



For students who are struggling to get started with this activity, consider prompting them with a low-entry question such as what they notice about the graph $h(x)$ and what they notice about the graph $d(x)$. Then help connect them back to question 1.



Consider allowing students to use the parent functions with sliders in Desmos to visually see the shift.

13.6.2 Exploration: Vertical Transformations (continued)

3. The table of values for $y = d(x)$, $y = h(x)$, and $y = k(x)$ are shown.

$y = d(x)$		$y = h(x)$		$y = k(x)$	
x	y	x	y	x	y
-2	4	-2	9	-2	1
0	0	0	5	0	-3
2	4	2	9	2	1

Discuss with your partner how the shift in the graph is shown in the y -values of the tables.

To go from $d(x)$ to $h(x)$ the y -values increased by 5. To go from $d(x)$ to $k(x)$ the y -values decreased by 3.

4. Complete the statements.

- a. The graph of $y = h(x)$ is a
☐ horizontal shift
☒ vertical shift
 of the graph of $y = d(x)$.
- b. For the graph of $y = d(x)$ to coincide with the graph of $y = h(x)$, it is shifted
☐ down
☐ to the left
☐ to the right
☒ up
 5 units.
- c. The shift can be seen in the table of values by
☒ adding
☐ subtracting
 to/from each
☐ x -value
☒ y -value
 the function $d(x)$ for each corresponding
☐ x -value
☒ y -value
 shown in the table.
- d. The graph of $y = k(x)$ is a
☐ horizontal shift
☒ vertical shift
 of the graph of $y = d(x)$.



When circulating during this activity, listen for how students are responding to question 3. Students may have a misconception of where the shift is occurring in the table of values.



When the activity is complete, consider using question 4 as a discussion with the class. Connect these statements to the ones they made about horizontal shifts in the prior lesson, notating the distinct differences.



What are some things you notice about a graph that has shifted vertically compared to a graph that has been shifted horizontally?

13.6.2 Exploration: Vertical Transformations (continued)

e. For the graph of $y = d(x)$ to coincide with the graph

of $y = k(x)$, it is shifted

- ☐ down
- ☐ to the left
- ☐ to the right
- ☐ up

3 units.

f. The shift can be seen in the table of values by

- ☐ adding
- ☐ subtracting

to/from each

- ☐ x -value
- ☐ y -value.

the

function $d(x)$ for each corresponding
shown in the table.

- ☐ x -value
- ☐ y -value

5. Write the function for each shift in terms of $d(x)$.

Up 5 units	$h(x) = d(x) + 5$
Down 3 units	$k(x) = d(x) - 3$

6. Ask another student how they wrote the shift in terms of $d(x)$. Ask them to justify their answer and write a summary of their justification. *You are the only person who should write in the middle column of the table.* Have the person initial next to your summary stating that your summary is accurate.

Shift	My Summary of Their Justification	Initials
Up 5 units	<i>Answers will vary.</i>	
Down 3 units	<i>Answers will vary.</i>	

7. Review the information from other students with your partner and if necessary, change the functions for each shift.

Answers will vary. All answers should match to $h(x) = d(x) + 5$ and $k(x) = d(x) - 3$.



What is the relationship between the sign of the k -value and the direction of the vertical shift? How does that differ from a horizontal shift?



To add student movement and increase exposure to different viewpoints, consider having students talk to two different students that they have not worked with yet in this unit to complete the task and gather their information and initials.

13.6.3 Bring It Together: Vertical and Horizontal Transformations

Component Overview: Students will complete a table where they describe the effect on the graph and table of values for each function provided.



13.6.3 Bring it Together: Vertical and Horizontal Transformations

- Describe the effect on the graph and table of values of $f(x)$ for each.

Function Notation	k -values	Graph	Table of Values
$f(x + k)$	$k < 0$	shifts right	add to the x -values
$f(x + k)$	$k > 0$	shifts left	subtract from the x -values
$f(x) + k$	$k < 0$	shifts down	subtract from the y -values
$f(x) + k$	$k > 0$	shifts up	add to the y -values



Students may get confused by the column header “graph.” Consider providing additional instructions to remind students they are only listing the effect on the graph, such as “shifts down 3 units” in that cell. The same is true for the table of values. Some students will try to create a table of values, which is not the intent of that column.

13.6.4 Guided Practice: Describing Transformations

Component Overview: Students will complete four practice problems about transformations of different function types.



13.6.4 Guided Practice: Describing Transformations

- Complete the statement to describe how the graph of $g(x) = f(x) + 6$ compares to the graph of $f(x) = |x + 8|$.

The graph of $g(x)$ represents a
☐ horizontal
☒ vertical
 shift of

the graph of $f(x)$. The graph of $g(x)$ is shifted 6

units
☐ down from
☒ up from
☐ to the right of
☐ to the left of
 the graph of $f(x)$.

- Describe the effect on the x -values of $H(x) = |x + 2|$ for the transformation $K(x) = H(x - 7)$.
All the x -values would have an increase of 7 units.
- Describe the effect on the y -values of $w(x) = \frac{2}{5}x - 9$ for the transformation $m(x) = w(x) + 5$.
The y -values have an increase of 5 units.
- Describe how the graph of $P(x) = R(x + 11)$ compares to the graph of $R(x) = x^2 + 5$.
The graph shifts 11 units to the left from $R(x)$.



For questions 2-4, consider working with small groups of students who struggled in the prior activities, including the prior lesson with horizontal shifts, to make sure they are able to have their misconceptions addressed and that they understand the concept.



Teachers may want to have students graph $f(x)$ in Desmos using function, then using function notation in Desmos graph $f(x) + 6$.

13.6.5 Wrap-Up: Growth Mindset

Component Overview: Students will complete a Growth Mindset reflection activity.



13.6.5 Wrap-Up: Growth Mindset

1. Complete the table.

Answers will vary.

Which mindset did I demonstrate?	Check the best description.
Did I use the partner discussions as learning opportunities?	☐ Never ☐ Sometimes ☐ Usually ☐ Always
Did I work on problems that challenged me?	☐ Never ☐ Sometimes ☐ Usually ☐ Always
Did I use strategies to 'un-stick' myself when I found the problems difficult?	☐ Never ☐ Sometimes ☐ Usually ☐ Always



Consider our online resources for additional enrichment opportunities for students to practice or test their abilities.